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Drugs affecting the respiratory system

시험 : 중요한 내용 쉽게~

한 시간 안에 많은 이야기를 할 수 없음
asthma, COPD 중심으로 (환자가 많음)

약리학교실

이 성 용

학습목표

- asthma와 COPD 병태생리를 안다
- 기관지 확장제와 항염증제 작용기전과 임상적 작용 이해한다
- asthma stepwise therapy의 근거를 설명할 수 있다
- 호흡억제/자극 약물의 임상적 사용 한계를 안다

Diseases of the Respiratory System

- **Asthma**

- inflammation & hypersensitivity
- anti-inflammatory agents & bronchodilators

chronic 진행 시 염증이 문제

- **Hypersensitivity Pneumonitis & Pulm. infiltrates with eosinophilia**

- **Occupational & Environmental Lung Disease**

- exposure to toxic chemicals & dusts

- **Pneumonia**

- infected with pathogens & ventilator-associated
- antibiotics & support care

- **Cystic fibrosis**

- Interstitial lung disease**

- steroid, cyclophosphamide, azathioprine

- **COPD**

- tx. very similar with asthma

호흡기에 작용하는 약물들

I. Respiratory stimulant (호흡자극제)

호흡자극제 사용이 그렇게 흔한 경우는 아님
호흡자극제를 사용할 정도면 자가 호흡이 어려움

- used only in certain specific conditions

: *post-op. respiratory depression*

acute respiratory failure (e.g. severe apneic episodes)

overdose of sedatives (e.g. barbiturate) barbiturate 가장 큰 단점 : 강력한 sedation

- drug : **doxapram (Dopram)**

중추 호흡 중추 자극

2시간 내에 호흡 호전이 되어야 함

- stimulate carotid chemoreceptors (cf. CO₂) & respiratory center

- infusion must be finished within 2 hr (0.3 ~ 3 mg/kg/min)

- replaced by ventilator support

- contraindication

severe acute asthma, epilepsy, IHD → epilepsy : 호흡 근육을 포함한 골격근 수축
→ 그다지 도움이 되지 않음

drug overdose-induced resp. depression

almitrine 잘 사용하지 않지만 퇴출 × (교과서에 언급)

- piperazine derivate

- stimulate peripheral chemical receptor

- produce neuropathy by chronic use – limited in use

naloxone 치료 목적 아편유사제 사용 시 antidote

- against opioid-induced resp. depression

II. Drugs causing respiratory depression (호흡억제 약물)

임상적 목적 사용 X
호흡억제는 약물 유해효과로서 주의

CNS를 억제하기에 바람직하지 않은 adverse effect

- drugs can induce CNS depression result in resp. depression

- by overdose

: narcotic analgesics, barbiturates

barbiturate : 강력한 진정수면제, 마취제

but, 호흡억제 부작용 → benzodiazepine에 자리를 빼앗김

many H₁ histamine-receptor antagonists

ethanol

- by therapeutic dose range

: opiates

- dihydrocodeine & diazepam : antitussive & resp. depressant

barbiturate 단점 극복

- benzodiazepine cause little depression

진정수면제 + ethanol

→ CNS depression이 심해지므로 주의

may cause dangerous degree if combined with ethanol

- be careful to COPD or asthma patients

Bronchial Asthma

- def.: recurrent reversible airway obstruction structure damage, irreversible change까지 오지 않음
in response to wide range of triggers (stimuli)
일부 chronic 환자에서는 irreversible

cf: some of chronic asthma patients

& obstructive airway disease – not reversible

- sx: intermittent attacks of dyspnea(SOB), chest tightness, 호기가 잘 되지 않음

청진을 하지 않아도 들림 wheezing & cough

- features chronic 상태가 되면 inflammation 중요

inflammatory changes in the airways

associated with bronchial hypersensitivity

(bronchoconstriction)

- pathology

- epithelium, reduced attachment to airway lumen 내로 epithelium 탈락

: in the lumen

infiltration of eosinophils, T_H2 lymphocytes Th2 lymphocyte : 각종 cytokine 분비
(1) B-cell immunoglobulin 생성 증가
(2) eosinophil, mast cell, macrophage 등의 activation

- airway, thickening (basement memb, sm. M) & edematous

- airway, obstruction by mucous plug from goblet cell

plasma protein from ↑ blood vessel

Chronic Obstructive Pulmonary Diseases

- Types (actually, mixed) 환자 입장에서 세 가지 소견이 섞여 있음
(교과서적으로 세 가지 타입으로 분류)
 - emphysema surfactant ↓ → alveoli 탄력성 ↓ → gas exchange ↓
: destruction of memb. & enlargement of alveolar sac
loss of elastic recoil, air trapping
gas exchange surface ↓
 - chronic bronchitis
: chronic productive cough for ≥ 3 months in 2 years
sputum이 배출되는 cough
 - small airway disease
: small bronchioles narrowing, early COPD begins
in small airway (< 2 mm)
- SX : chronic cough & sputum production
wheezing, tightness feeling in the chest
constant fatigue, difficulty sleeping, morning headaches
weight loss

Chronic Obstructive Pulmonary Diseases



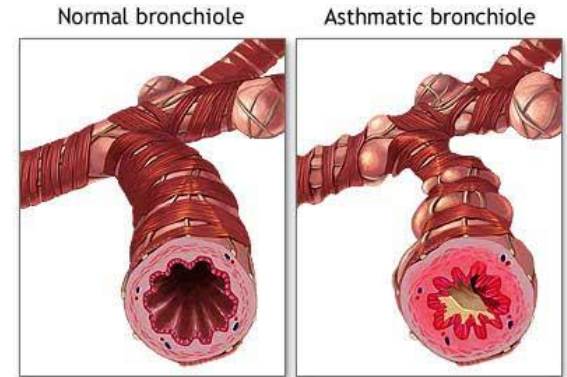
Asthma vs COPD

Asthma	COPD
일부는 irreversible reversible	irreversible
과민반응의 nature → 호산구 dominant eosinophilic	과민반응은 아님 neutrophilic
핵심적인 치료제 ICS 중요	LABA/LAMA 중요
onset young	smoking-related

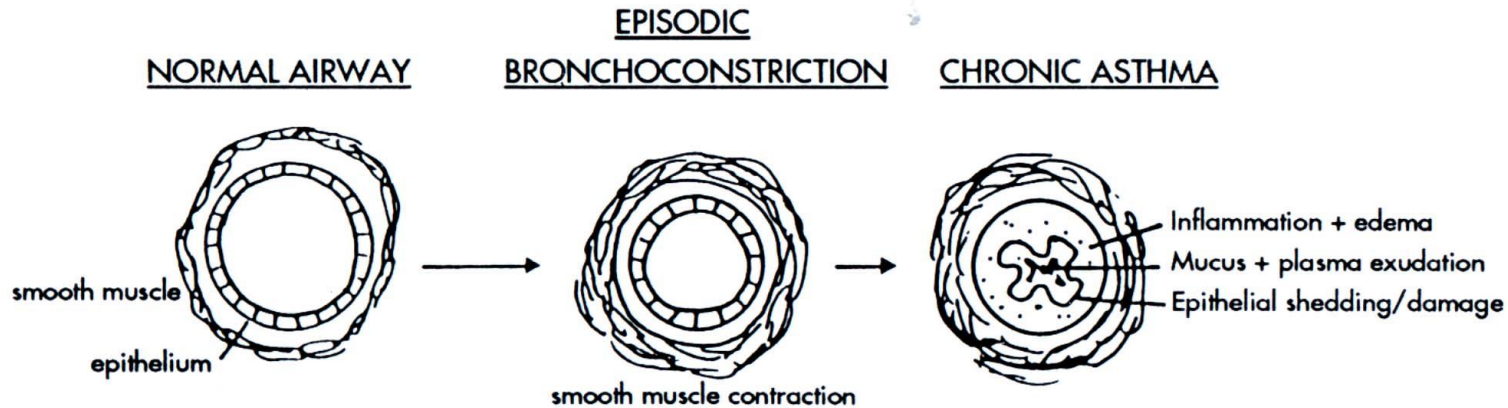
tirgger에 의한 bronchoconstriction → 이완제가 도움이 됨
 intermittent attack 반복 → 염증을 피할 수 없음 → anti-inflammatory drug

천식 치료제

분류와 제제



ASTHMA THERAPY



BRONCHODILATORS

β_2 Agonist
 Theophylline
 Anticholinergic

5~10년 전에는 많이 사용했으나 중요도 감소

ANTIINFLAMMATORY DRUGS

Corticosteroid
 Cromolyn sodium 예방적 차원에서 사용
 Nedocromil sodium
 Ketotifen
 Methotrexate?
 Gold??

Aims of asthma therapy 대증요법이 주요 치료

- minimal chronic symptoms, including nocturnal 수면질 저하 방지
- minimal exacerbations 악화 방지
- prevent reattack
- no emergency visit
- no limitations on activities physical activity도 일종의 trigger
직업을 잃게 될 수도 있음(특히, 운동선수는 치명적)
- nearly normal peak expiratory flow 천식 환자는 호기가 어려움
→ 1초 내 힘차게 호기 시 volume이 매우 떨어져 있음
- minimal adverse effects from medicine

Bronchodilator (기관지 확장제)

- β_2 -(adrenoceptor) agonists : short-acting (SA) & long-acting (LA)
- methylxanthines : long-acting bronchodilator
- anticholinergics (muscarinic antagonists)
- others :

short-acting bronchodilators - for relief of bronchoconstriction

e.g. epinephrine / ephedrine

long-acting bronchodilators - for prevention

attack 순간 모면보다 예방적 목적이 큼
(공장, 좋지 않은 냄새 등의 주변 환경에 속한 경우)

Bronchodilator

β_2 -(adrenoceptor) agonists

cf. β_1 : mostly in heart

- action mechanism

- **relax airway smooth muscle cells** 교감신경 흥분 → 기관지 평활근 이완 (via β receptors in airway)

: $\uparrow$ cAMP by activation of adenylate cyclase

- inhibit 'mediator release' from **mast cells** & monocytes

각종 chemical mediator 분비

→ inhibits inflammation

- preparations for administration 다양한 제제

aerosol, powder, oral, or injection

Bronchodilator

β_2 -(adrenoceptor) agonists

- two categories

(1) short-acting β_2 agonists (SABAs)

albuterol & terbutaline 

attack이 왔을 때 도움

rapid onset of bronchodilation: 'rescue drug' begin (5-10 min),

max effect within 30 min, action duration: 3-6 hr

몇 시간 지속되지 않음
but, attack trigger에서 벗어날 수 있음

as needed for sx. relief, prevent EIA (taken prior to exercise)

(2) long-acting β_2 agonists (LABAs)

salmeterol & formoterol 

action duration : >12 h, given twice daily

should be combined with ICS LABA 단독 보다 ICS 병용 투여 권유

Effects of beta-adrenergic agonists on airway

- relaxation of airway smooth muscle
 - most potent bronchodilator
 - inhibition of mast cell mediator release
 - inhibition of plasma exudation and airway edema
 - increased mucociliary clearance 어느 정도 내경 확보 가능
 - decreased mucus secretion
 - decreased cough
 - no effect on chronic inflammation not anti-inflammatory drug
→ 단독으로 항염증 작용을 기대하기 어려움
- aerosol type, esp. against mild asthma

Side effects & Tolerance of beta-adrenergic agonists

aerosol 형태는 부작용이 없어서 장기간 사용에 문제 없음

- almost little S/E in inhalation administration
- esp. in elderly patients 노인에서는 교감신경 흥분 양상으로 나타남
 - muscle tremor distal extensor, flexor 반복 수축
 - palpitation
- insomnia, hypokalemia, arrhythmia
- chronic use, induces tolerance
 - : not beta receptor issue
 - “mast cell” tolerance
 - prevented by inhaled corticosteroid

Methylxanthines

몇 년 전까지는 많이 사용했으나 부작용으로 인해 사용 감소
but, 여전히 처방하기는 함

교감신경 흥분 작용 + 기관지 확장 작용

- **theophylline**, theobromine & caffeine (oral bronchodilator)

doxofylline – less S/E

(IV) **aminophylline** (soluble salt of theophylline): less potent && shorter
recommended dosage: 8-14 µg/dL

- actions

- has narrow therapeutic index
- **inhibition of phosphodiesterase (cAMP ↑) in airway smooth m.**
- **lower dose has anti-inflammatory effect** 어느 정도의 항염증 작용

: activates key nuclear enzyme **histone deacetylase-2**

(inactivate inflammatory gene)

- indications

앞서 언급한 표에서도 COPD는 LABA 중요

only in chronic obstructive pulmonary disease (COPD)

bronchial asthma, rarely used

CNS 흥분 부작용

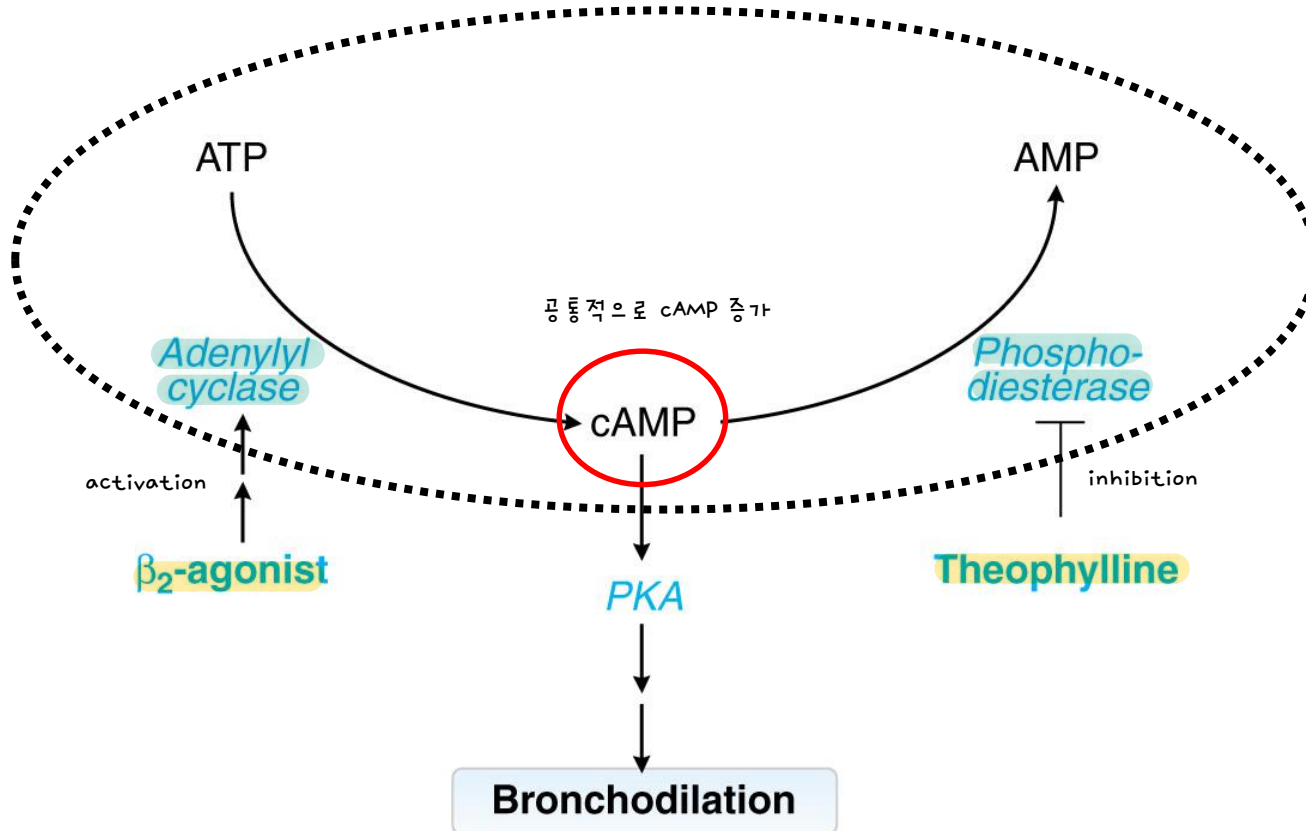
seizure를 앓고 있거나 과거력이 있다면 피해야 함

- S/E : N/V, headache, diarrhea, CNS activation-related sx.s (seizure)

diuresis, palpitation, arrhythmia

간접적인 이뇨작용

Beta-agonist와 theophylline의 기관지확장 효과



Anti-cholinergics (muscarinic receptor antagonist) choline influence를 줄여 호흡에 도움

- *ipratropium*: (trade name **Atrovent**)
 - derivative of N-isopropyl-atropine
 - main compound as an anti-asthmatic
- tiotropium*
- relax bronchoconstriction & reduce mucus secretion
 - by parasympathetic stimulation
- *no effect on the late inflammatory phase of asthma*
- combined with **salbutamol** for the management of COPD & asthma
 - trade name: **Combivent**, (tx for obstructive lung disease) 병합투여 약물
- aerosol inhalation, max action within 30 min, lasts 3-5 h
- safe & well tolerated 안전하고 특별한 유해효과 없음
 - anticholinergic side effect, minimal
 - dry mouth & sedation have been reported
 - urinary retention BPH 환자라면 영향을 받을 수 있음

NORMAL

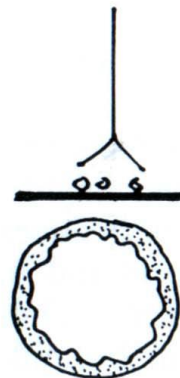


COPD

흡연도 원인이지만 높아진 vagal tone도 원인
→ vagal tone을 줄이기 위해 항콜린 약물 사용



Vagal "tone" $\propto$ 10 : 부교감 신경



Cholinergic antagonist

Resistance $\propto 1 / \text{radius}^4$

Epinephrine

- effective rapid acting bronchodilator
강한 제제이므로 희석 용액을 subcu 주사
- injected subcutaneously, 0.4 ml of 1:1,000 solution
- inhaled as a microaerosol, 320 mcg per puff
- max. bronchodilation achieved in 15 min after inhalation & lasts
for 60-90 min
- used in anaphylaxis
- S/E: tachycardia, arrhythmia, worsening of angina
arrhythmia 과거력, tendency가 있다면 조심
ionotropic action 증가 → myocardium oxygen demand 증가 → angina 악화

Ephedrine

- alkaloid obtained from ephedra
- acts on both alpha & beta receptors
- has long duration of action
- produces mild stimulation of CNS 적정 용량 사용 시
- now infrequently used for asthma asthma 치료제로 항상 소개하긴 함



Isoproterenol

- acts on beta₁ & ₂ receptors
- *potent bronchodilator* when inhaled 흡입 시 강한 작용
- 80-120 mcg produces max bronchodilator within 5 min
- has 60-90 min duration of action 짧은 action duration
- S/E: arrhythmias 교감신경 흥분 관련이므로 arrhythmia 조심

some psychostimulant drugs 바람직한 약물은 아니지만 이론적으로 bronchodilation

- have an amphetamine like mode of action

such as *amphetamine*, *methamphetamine*, & *cocaine*

→ have bronchodilating effects

Anti-inflammatory agents (항염증제)



- corticosteroids
- anti-leukotrienes
- lipoxygenase inhibitor
- mast cell stabilizer
- anti-IgE antibody

Anti-inflammatory agents (항염증제)



Corticosteroids (inhaled, ICS)

- not bronchodilator

for management of chronic asthma with predominant inflammatory component

→ most effective controllers for asthma

- can prevent EIA & nocturnal exacerbation 예방 가능

early tx : prevent irreversible change of airway fx in chronic asthma

일반적으로 COPD와 달리 irreversible change가 나타나지는 않음

- actions 각종 염증반응 억제 mechanism

: most effective antiinflammatory agents for asthma tx

- decrease inflammatory cell number in airway

(eosinophils, T cell, mast cell)

- inhibit allergen-induced influx of eosinophils into the lung

- decrease cytokine production (from eosinophil)

← switch off the activated gene encoding inflammatory protein

- inhibit the production of vasodilator, PGE_2 and PGI_2 (**COX2**) COX2 억제 → PG 효과 억제

LTC_4 , LTD_4 , & LTB_4 & PAF 강력한 bronchodilation 매개체 : leukotriene

- activate the antiinflammatory genes such as MAP kinase phosphatase-1

- decrease microvascular permeability 염증 반응의 근간

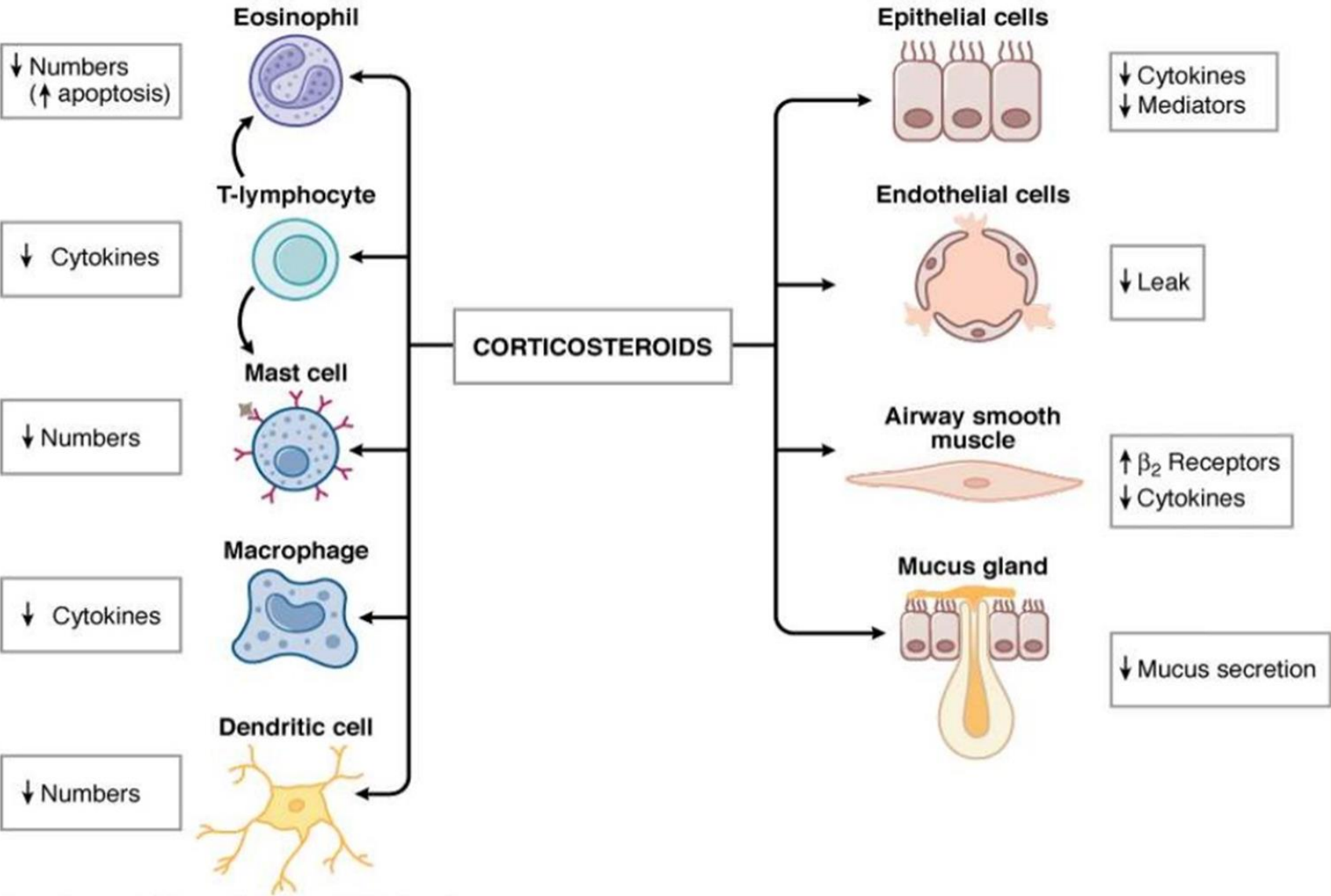
allergy, inflammation 공통 반응 : 혈관 확장

각종 염증세포 역할이 무너지짐

INFLAMMATORY CELLS

structural cell에도 많은 작용

STRUCTURAL CELLS



Corticosteroids (inhaled, ICS)

45

- compounds

beclomethasone, triamcinolone, budesonide, fluticasone

cf. hydrocortisone : IV for status asthmaticus → hydrocortisone 사용
intermittent attack은 그나마 견딜만 함
status asthmaticus는 환자가 고통스러움

- clinical use

~ *not cure underlying condition* (withdrawal sx +)

usually given twice daily

beneficial in treating asthma of any severity & age

rapidly improve asthma sx. & lung fx. (in several days)

preventing EIA, nocturnal exacerbations, & severe exacerbations

maximal improvement needs tx for several months

- S/E

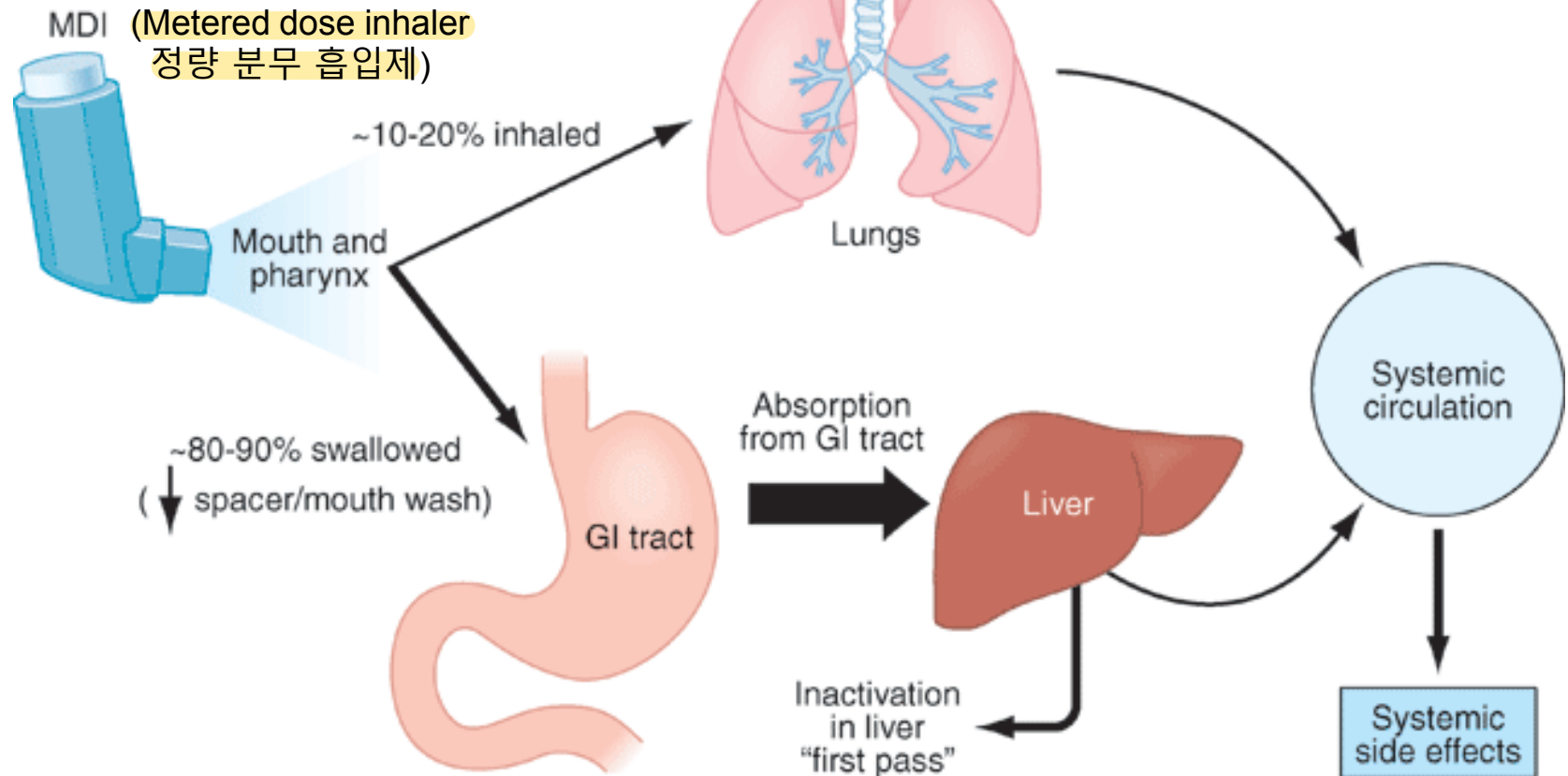
ICS, minimal systemic S/E

hoarseness & weakness of voice (dysphonia) 환자에게 목 소리가 변할 수도 있음을 미리 알려야 함

← atrophy of the vocal cord by laryngeal deposition of steroid
oral candidiasis

Pharmacokinetics of inhaled corticosteroids

환자 증상에 따라 처방 → 환자마다 한 번 누를 때마다 분무되는 양이 다름
but, 가성비(효율)가 좋지 않음 why? 90%는 GI로 가고 10%만 효과



흡입형 ICS는 장기간 사용해도 위험하지 않음

비만세포 탈과립 억제제 (Mast cell stabilizer)

알러지 반응, 염증 반응이 국한된 위치에 나타나는 이유
: mast cell 분포

Cromolyn (disodium cromoglycate) 소아 환자에게 많이 사용

- not have any direct effect on smooth muscle
- preventing histamine release from mast cell
cytokine from inflammatory cells
- used **prophylactically** to reduce both immediate & late-phase
asthmatic responses
- 이론적 - effective in Ag-, exercise-, and irritant-induced asthma
- preparation: *nasal spray, oral medication (food allergy), eye drop*
- children, more likely to respond than adult, 4 times a day
- S/E : few, cough, airway irritation

Nedocromil

- inhibits the degranulation of mast cells, prevents release of histamine, so preventing the synthesis of PGs & LTs.

Anti-leukotrienes (bronchodilation & anti-inflammatory effects)

- **cysteinyl-leukotriene receptor inhibitors**

- **cysteinyl leukotriene LTC₄, LTD₄ & LTE₄**

potent chemical mediator (mainly from mast cell)

→ induce smooth muscle constriction (100 times than histamine)
천식 환자가 항히스타민제를 사용하는 것과 동일

induce inflammation process (microvascular leakage)

- less effective than ICS

- **montelukast & zafirlukast**

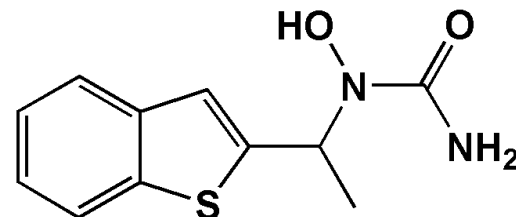
👁👁 *only for prophylactic use*, not for tx of acute asthma attack

S/E: headache, dizziness, abdominal pain, sore throat

- **zileuton** : **5-lipoxygenase inhibitor**, inhibits LTB₄ ~ E₄ productions
prophylactic use & chronic asthma in **adult & children >12 yr**

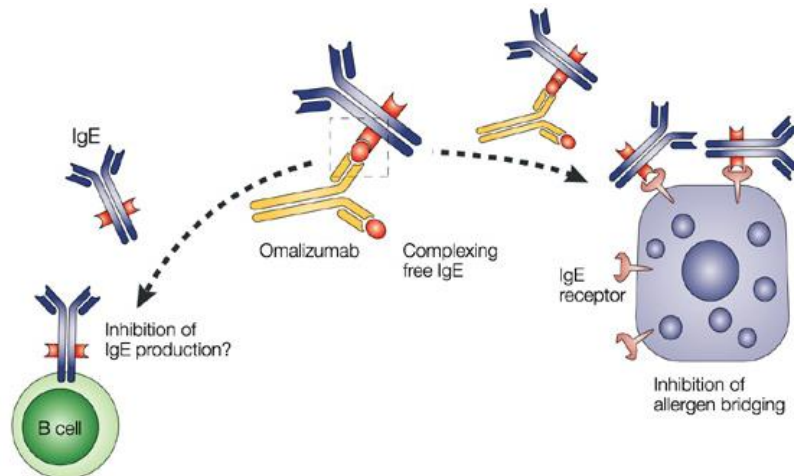
contraindicated in liver diseased state

S/E – sinusitis & nausea, sleep disorder



Anti-IgE antibody

- **omalizumab** monoclonal Ab
 - blocking Ab
 - **neutralizes circulating IgE** (bind to CR3 region)
 - inhibits IgE-mediated reactions
 - reduces the number of exacerbations in patients in severe asthma who are poorly controlled with conventional therapy
 - approved in 2003, **very expensive** 단백질 제제는 고가
 - 3-4 month trial of therapy
 - subcutaneous injection, every 2-4 weeks



Stepwise Approach for Managing Asthma

(Age ≥ 12 years)

severity 구분하여 약물 결정

Intermittent Asthma

mild, intermittent

Step 1

Preferred:
SABA PRN

Persistent Asthma: Daily Medication

Consult with asthma specialist if step 4 care or higher is required.
Consider consultation at step 3.

step이 올라갈수록 약물 종류, 강도 증가

Step 2

Preferred:
Low-dose ICS

Alternative:
Cromolyn,
LTRA,
Nedocromil, or
Thophylline

Step 3

Preferred:
Low-dose ICS +
LABA
OR
Medium-dose
ICS

Alternative:
Low-dose ICS +
either LTRA,
Thophylline or
Zileuton

Step 4

Preferred:
Medium-dose
ICS + LABA

Alternative:
Medium-dose
ICS +
either LTRA,
Thophylline or
Zileuton

Step 5

Preferred:
High-dose ICS
+ LABA

AND

Consider
Omalizumab
for patients
who have
allergies

Step 6

Preferred:
High-dose ICS
+ LABA + oral
corticosteroid

AND

Consider
Omalizumab
for patients
who have
allergies

Step up if needed

first check
adherence,
environmental
control and
comorbid
conditions

Assess Control

Step down if possible

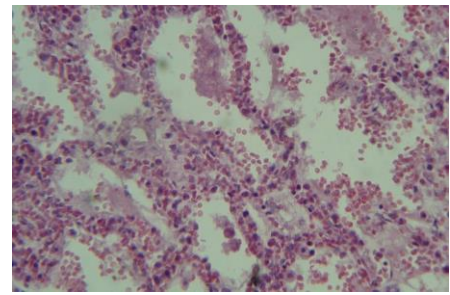
and asthma
is well
controlled
at least
3 months

Each step: Patient education, environmental control, and management of comorbidities.

Quick-Relief Medication for All Patients

- SABA as needed for symptoms. Intensity of treatment depends on severity of symptoms: up to 3 treatments at 20-minute intervals as needed. Short course of oral systemic corticosteroids may be needed.
- Use of SABA >2 days a week for symptom relief generally indicates inadequate control and the need to step up treatment.

Adult respiratory distress syndrome (ARDS) 우리 나라에서 흔 하지 는 않음

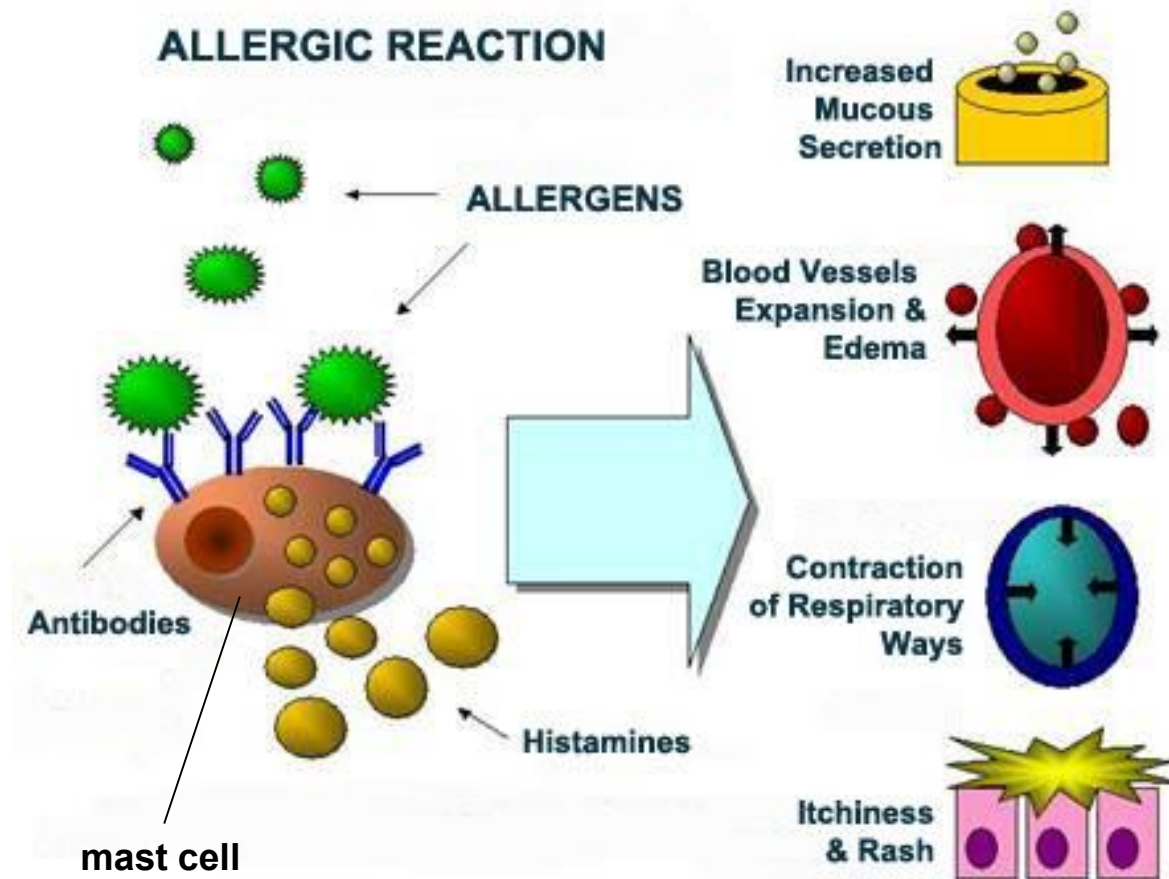


- 60% mortality
- pulmonary arterial hypertension with vasoconstriction
- extensive injury and occlusion of the pulmonary microvasculature
 - edema & progressive hypoxia gas change ↓
multi-organ dysfunction occur
- etiology
 - sepsis (M/C), pneumonia, aspiration of gastric contents, toxic chemicals, drug reaction, trauma
- pathology
 - inflammatory mediators are involved
 - diffuse inflammation of lung parenchyma
diffuse alveolar damage & hyaline memb. in alveolar walls
 - result in sepsis via lung infection
- tx: inhaled nitric oxide (as selective pulmonary vasodilator)
corticosteroid

Allergic rhinitis 높은 histamine dependency → 항히스타민제 사용

- hay fever, nasal allergies
- sx: sneezing, itchy nose/eyes
watery rhinorrhea, nasal congestion
- when, allergic to dust, dander, insect venom, or pollen (allergen)
→ body release chemicals including histamine

Role of histamine in allergic reaction



Clinical uses of anti-histamine

histamine이 pathophysiology에 관여하는 질병

- Allergic rhinitis (common cold)
- Allergic conjunctivitis (pink eye)
- Allergic dermatological conditions
- Urticaria (hives)
- Angioedema (swelling of the skin)
- Pruritus (atopic dermatitis, insect bites)
- Anaphylactic reactions (severe allergies)
- Nausea and vomiting (first generation H1-antihistamines)
- Sedation (first generation H1-antihistamines)

Corticosteroids

These prescription sprays reduce inflammation of the nose and help relieve sneezing, itching, and runny nose. It may take a few days to a week to see improvement in symptoms.

Beclomethasone

Fluticasone

Mometasone

Triacinolone

Drug used for cough (antitussives)

- cough : protective reflex mechanism, to remove foreign materials & secretions from bronchi & bronchioles
can be stimulated by inflammation in the resp. tract or neoplasm
- antitussive (cough suppressant) sometimes used for dry painful cough associated with carcinoma & inflammation
→ merely suppress the sx.
not used for the cough associated with asthma
narcotic analgesics
- **Opiate (Codeine)** – moderate agonist of mu-receptor
 - 3-methylmorphine, less addictive
 - effective cough depressant
 - decrease secretion from bronchioles
 - euphoria, itching, N/V, urinary retention, constipation, miosis
- **Dextromethorphan**
 - synthetic narcotic analgesic
 - potency similar to codeine, less constipation
 - NMDA, opioid receptor inhibitor

두 가지 중요

Drug used for cough (antitussives)

- Benzoate
 - local anesthetic
 - inhibit the stretch receptor in resp. tract, lung, pleura
 - S/E : dizziness, convulsion, cardiac arrest
 - para-aminobenzoic acid : metabolite, may produce allergy